Operating system and system programming (25CS2104E)

ALARM CLOCK

Review-I

Under the Esteemed Guidance of

Raghupathi M
CSE Department

Team Members:
M.Priyavallika - 2520090183
D.Shamika Hasini - 2520080033
Samyuktha Kandukuri - 2520090157

Contents

Abstract

The Alarm Clock project is a Linux-based system designed to provide a simple and reliable way to schedule and trigger alarms at a specified time. The application monitors the system clock and notifies the user when the scheduled alarm time is reached. The project mainly focuses on demonstrating Operating Systems and system programming concepts through Linux/POSIX mechanisms.

Key features and concepts include:

- Setting, viewing, and cancelling alarms.
- Monitoring the current system time.
- Triggering an alarm at the scheduled time.
- Using Linux processes and signals for alarm management.
- Using POSIX timers and time-related system calls.
- Providing a simple command-line interface for user interaction.
- Demonstrating practical Operating Systems concepts such as process management, signal handling, and time-based scheduling.

The expected outcome is a functional Linux-based Alarm Clock that performs its intended task while providing practical experience with Linux system calls, POSIX APIs, and Operating Systems concepts.

Introduction

An Alarm Clock is a time-based system that allows users to schedule an alarm for a specific time and receive a notification when the set time is reached. In this project, a Linux-based Alarm Clock is developed to demonstrate how Operating Systems and system programming concepts can be used to implement time-dependent tasks.

The system uses the Linux system clock to monitor the current time and manages the alarm using POSIX mechanisms. It focuses on concepts such as process management, signal handling, timers, and system calls. The application provides a simple interface through which users can set, view, and cancel alarms.

The main aspects of the project are:

- Scheduling alarms for a specified time.
- Monitoring the system time continuously.
- Triggering an alarm when the scheduled time is reached.
- Using Linux/POSIX APIs for timing and signal handling.
- Demonstrating practical Operating Systems and system programming concepts.
- Providing a simple and reliable Linux-based alarm solution.

Hardware and Software requirements

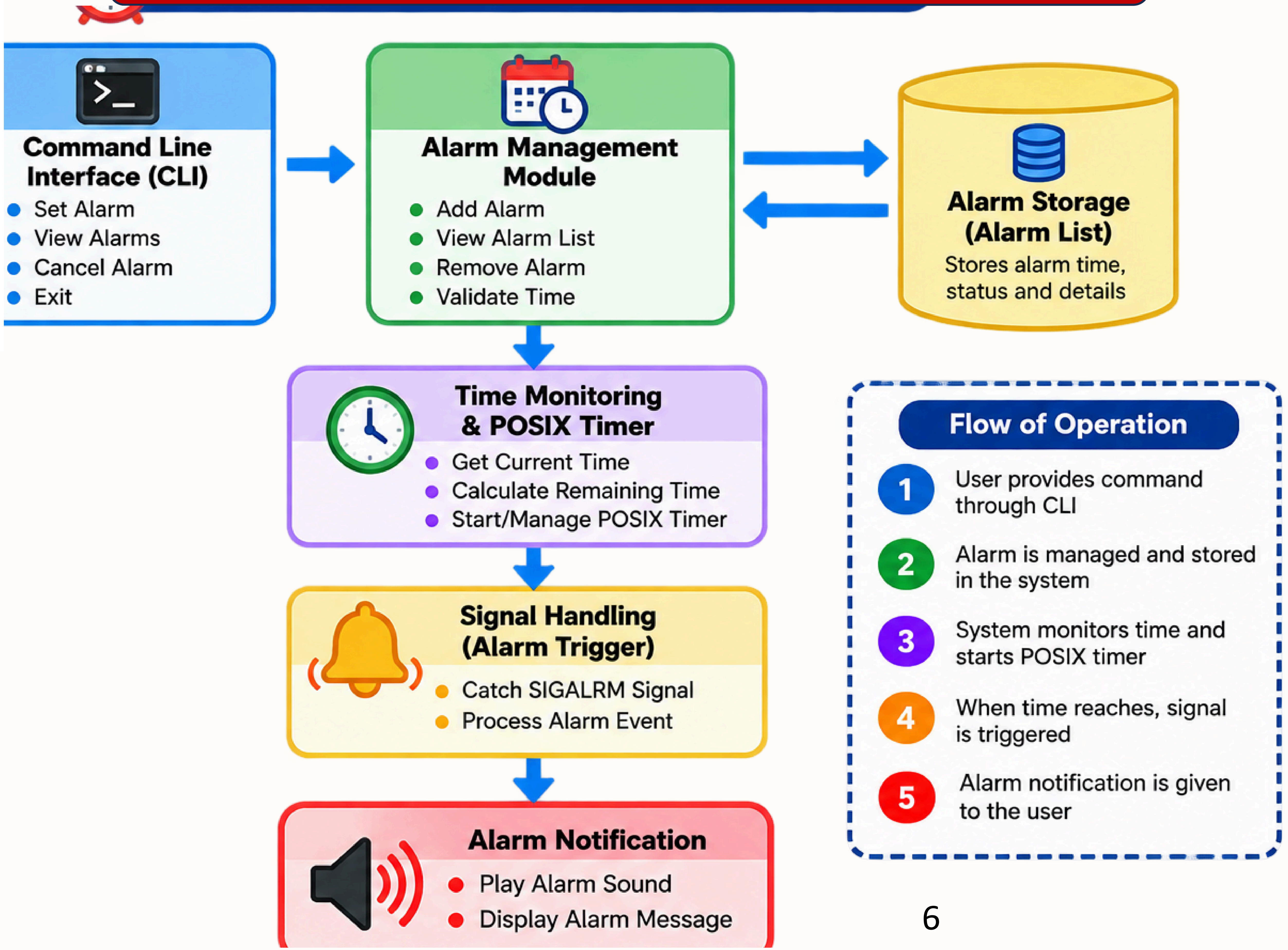
Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- 1 GHz or higher processor
- Speaker for alarm sound

Software Requirements

- Ubuntu/Linux operating system
- C/C++ programming language
- GCC compiler
- POSIX libraries
- VS Code or Vim editor

System Architecture



Thank you